

Smart Desk Automation System

Project Idea

The Smart Desk project is designed to automate desk interactions using sensors and actuators. It integrates a PIR motion sensor, a digital light sensor (LDR), a servo motor for drawer control, LEDs for ambient indication, and a 7-segment display for session timing. The system responds to user presence and ambient light conditions to control drawer access and lighting, enhancing convenience and energy efficiency. Inputs and Outputs

Inputs:

- clk: 50 MHz system clock
- rst_n: Active-low reset signal
- presence_raw: PIR sensor input (active-low, detects motion)
- btn_open: Manual drawer open button
- btn_close: Manual drawer close button
- light_dark: Digital LDR input (0 = dark, 1 = bright)

Outputs:

- led_on[9:0]: Controls 10 LEDs based on presence and light
- pwm_servo: PWM signal to control SG90 servo motor for drawer
- seg_units: 7-segment display for units digit of session timer
- seg_tens: 7-segment display for tens digit of session timer

Pin Assignment

The following table summarizes the pin assignments used in the project:

Signal	Direction	Pin	Voltage	Current
seq_tens[4]	Output	PIN M20	2.5 V	12mA (default)
seq_tens[5]	Output	PIN N19	2.5 V	12mA (default)
seq_tens[6]	Output	PIN N20	2.5 V	12mA (default)
rst_n	Input	PIN C10	2.5 V	12mA (default)
pwm_servo	Output	PIN V8	2.5 V	12mA (default)
presence_raw	Input	PIN AA14	2.5 V	12mA (default)
light_dark	Input	PIN W12	2.5 V	12mA (default)
led_on[0]	Output	PIN A8	2.5 V	12mA (default)
led_on[1]	Output	PIN A9	2.5 V	12mA (default)
led_on[2]	Output	PIN A10	2.5 V	12mA (default)
led_on[3]	Output	PIN B10	2.5 V	12mA (default)

led_on[4]	Output	PIN D13	2.5 V	12mA (default)
led_on[5]	Output	PIN C13	2.5 V	12mA (default)
led_on[6]	Output	PIN E14	2.5 V	12mA (default)
led_on[7]	Output	PIN D14	2.5 V	12mA (default)
led_on[8]	Output	PIN A11	2.5 V	12mA (default)
led_on[9]	Output	PIN B11	2.5 V	12mA (default)
clk	Input	PIN N14	2.5 V	12mA (default)
btn_open	Input	PIN B8	2.5 V	12mA (default)
btn_close	Input	PIN A7	2.5 V	12mA (default)

seq_tens[4]	Output	PIN M20	6	B6 N0	PIN M20	2.5 V	12mA (default)	2 (default)
seq_tens[5]	Output	PIN N19	6	B6 N0	PIN N19	2.5 V	12mA (default)	2 (default)
seq_tens[6]	Output	PIN N20	6	B6 N0	PIN N20	2.5 V	12mA (default)	2 (default)
rst_n	Input	PIN C10	7	B7 N0	PIN C10	2.5 V	12mA (default)	
pwm_servo	Output	PIN V8	3	B3 N0	PIN V8	2.5 V	12mA (default)	2 (default)
presence_raw	Input	PIN AA14	4	B4 N0	PIN AA14	2.5 V	12mA (default)	
light_dark	Input	PIN W12	4	B4 N0	PIN W12	2.5 V	12mA (default)	
led_on[0]	Output	PIN A8	7	B7 N0	PIN A8	2.5 V	12mA (default)	2 (default)
led_on[1]	Output	PIN A9	7	B7 N0	PIN A9	2.5 V	12mA (default)	2 (default)
led_on[2]	Output	PIN A10	7	B7 N0	PIN A10	2.5 V	12mA (default)	2 (default)
led_on[3]	Output	PIN B10	7	B7 N0	PIN B10	2.5 V	12mA (default)	2 (default)
led_on[4]	Output	PIN D13	7	B7 N0	PIN D13	2.5 V	12mA (default)	2 (default)
led_on[5]	Output	PIN C13	7	B7 N0	PIN C13	2.5 V	12mA (default)	2 (default)
led_on[6]	Output	PIN E14	7	B7 N0	PIN E14	2.5 V	12mA (default)	2 (default)
led_on[7]	Output	PIN D14	7	B7 N0	PIN D14	2.5 V	12mA (default)	2 (default)
led_on[8]	Output	PIN A11	7	B7 N0	PIN A11	2.5 V	12mA (default)	2 (default)
led_on[9]	Output	PIN B11	7	B7 N0	PIN B11	2.5 V	12mA (default)	2 (default)
clk	Input	PIN N14	6	B6 N0	PIN N14	2.5 V	12mA (default)	
btn_open	Input	PIN B8	7	B7 N0	PIN B8	2.5 V	12mA (default)	
btn_close	Input	PIN A7	7	B7 N0	PIN A7	2.5 V	12mA (default)	

Code Explanation

The VHDL code consists of several modules:

- PIR Synchronizer: Synchronizes the raw PIR input to the system clock.
- Tick Generator: Generates a 1-second tick using the 50 MHz clock.
- Button Pulse Generator: Detects rising edges of manual open/close buttons.
- Drawer Logic: Controls drawer state based on PIR input, buttons, and auto-close timer.
- LED Logic: Adjusts LED output based on presence and ambient light.
- Session Timer: Counts seconds of user presence up to 99 seconds.
- 7-Segment Decoder: Converts session counter digits to display format.
- Servo Driver: Generates PWM signal to control servo angle based on drawer state.

Results

- The drawer opens when motion is detected or the open button is pressed.
- The drawer closes automatically after 10 seconds or when the close button is pressed.
- LEDs turn on in darkness when a person is detected, and turn off otherwise.
- The session timer increments every second while a person is present.
- The servo motor rotates between 0° and 180° to open/close the drawer.

This system demonstrates a responsive and efficient smart desk interface using VHDL and FPGA-based control.